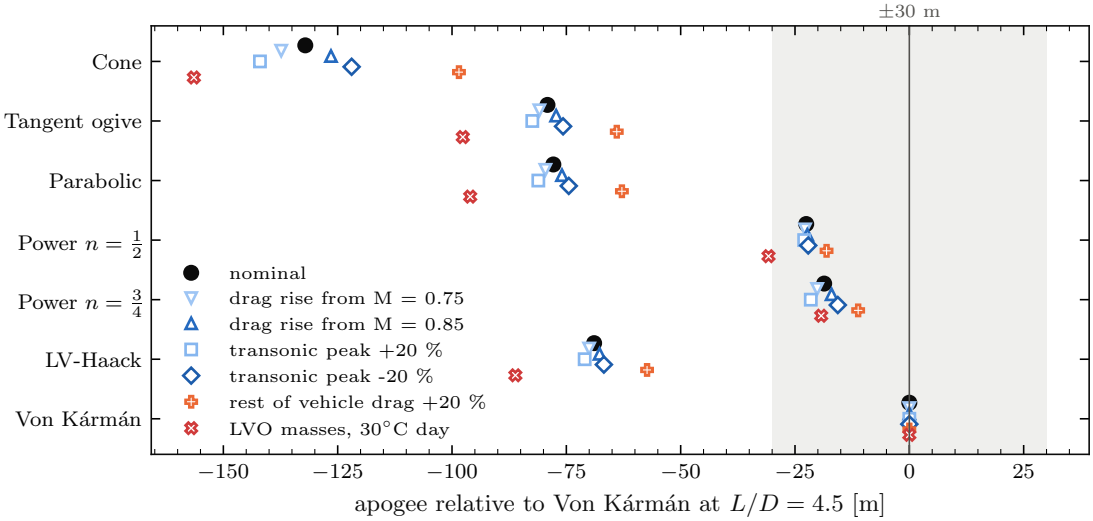

±30 m
Cone
Tangent ogive
Parabolic
Power $n = \frac{1}{2}$
Power $n = \frac{3}{4}$
LV-Haack
Von Kármán
nominal
drag rise from M = 0.75
drag rise from M = 0.85
transonic peak +20 %
transonic peak -20 %
rest of vehicle drag +20 %
LVO masses, 30°C day
−150
−125
−100
−75
−50
−25
0
25
apogee relative to Von Kármán at $L/D = 4.5$ [m]